

DAY 5 · SIMULATION

SITL & Missions

Fly the real firmware on your laptop. Plan a mission, execute it, and see the MAVLink underneath.

~15 minutes · then you fly your own mission in SITL



S I M U L A T I O N

SITL — fly without flying

Software In The Loop. The real ArduPilot firmware, running on your laptop instead of on a Pixhawk.
Same code, same parameters, no risk of crashing hardware.

Start the simulator — Mission Planner

Click **SIMULATION** in the top bar.



WHAT SITL IS

The **exact same firmware** that runs on your Pixhawk 6C, compiled for your laptop. It simulates the physics — gravity, drag, motor response — so the autopilot thinks it is flying a real aircraft.

WHY MISSION PLANNER

Mission Planner has a built-in SITL launcher. One click, no setup. It downloads the firmware, starts the simulation, and connects the GCS automatically.

Pick Multirotor — same as your real aircraft



SELECT THE VEHICLE

Click **Multirotor**. Set the home location by dragging the pin on the map — this is where the simulated aircraft spawns. Leave firmware as **Stable**.

SIM SPEED

Default is 1x. You can speed it up to test long missions faster, but start at 1x so the behaviour matches real flight.

The simulator connects to Mission Planner over TCP on localhost. The GCS cannot tell the difference between SITL and a real aircraft.

Switch to the Plan view — create a mission

Click **PLAN** in the top bar. The SITL aircraft is connected and ready.



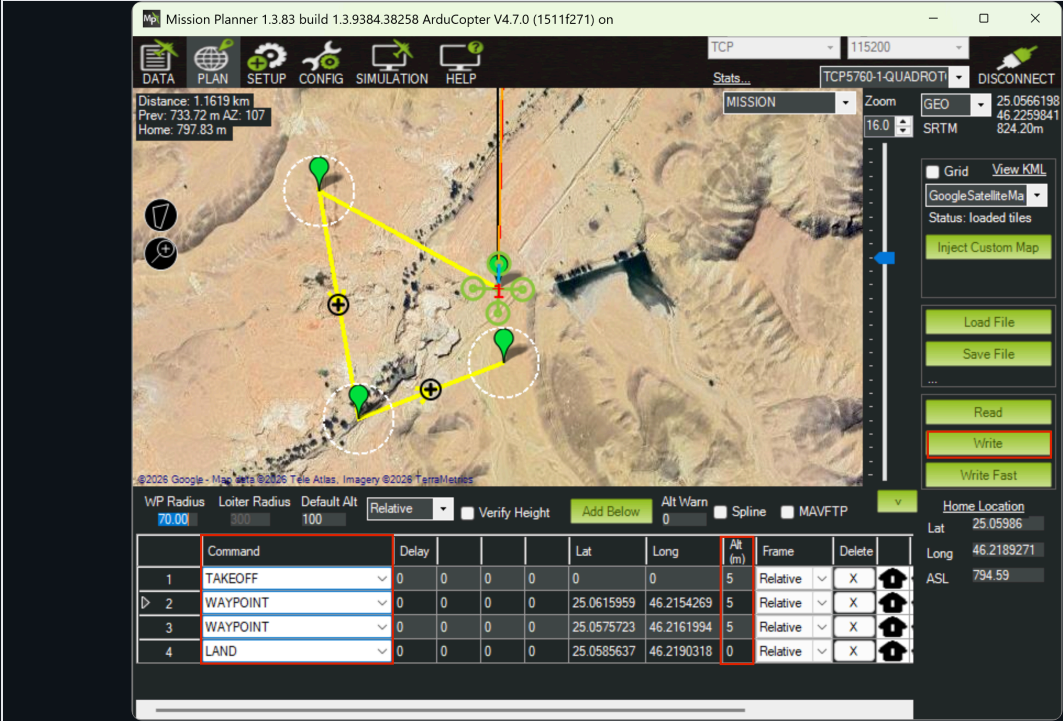
CONNECTED TO SITL

Notice the top-right shows **TCP5760-1-QUADROTOR** — that is the simulated vehicle. Battery 100%, GPS fix, Stabilize mode. Ready to plan.

PLAN VIEW

Click on the map to drop waypoints. Each click adds a mission item to the table at the bottom.

A complete mission — four items



THE MISSION TABLE

#	COMMAND	WHAT IT DOES
1	TAKEOFF	Climb to 5 m altitude
2	WAYPOINT	Fly to first point
3	WAYPOINT	Fly to second point
4	LAND	Descend and disarm

WRITE → ARM → AUTO

Click **Write** to upload the mission to SITL. Switch to the DATA screen, arm the vehicle, set mode to **Auto**, and watch it fly the mission.

What the mission looks like in MAVLink

Each row in the mission table is a MAVLink command. Take waypoint #2:

MAV_CMD_NAV_WAYPOINT (16)		
FIELD	VALUE	MEANING
Command	16	NAV_WAYPOINT
Param 1	0	Hold time (seconds)
Param 2	0	Acceptance radius (metres)
Param 3	0	Pass-through (0 = stop at WP)
Param 4	0	Desired yaw (NaN = unchanged)
Lat	25.0615959	Latitude
Lon	46.2154253	Longitude
Alt	5	Altitude (relative, metres)

EVERY MISSION ITEM IS THIS

TAKEOFF is command **22**. LAND is command **21**. WAYPOINT is command **16**. The GCS builds these from your clicks on the map and uploads them as a sequence.

Full list of NAV_ commands:
mavlink.io/en/messages/common.html

The mission you drew on the map is a list of MAVLink commands with coordinates. **That is all a mission is — numbered instructions the autopilot executes in order.**

SITL standalone — run it yourself

If you want SITL without Mission Planner — three commands in a terminal.

```
# 1. Clone the ArduPilot repo

git clone --recurse-submodules https://github.com/ArduPilot/ardupilot.git

cd ardupilot

# 2. Install the build environment

Tools/environment_install/install-prereqs-ubuntu.sh -y

# 3. Start the simulator

sim_vehicle.py -v ArduCopter --map --console
```

WHAT THIS GIVES YOU

A full SITL instance with a MAVProxy console and a map window. Connect QGC or Mission Planner to `tcp:127.0.0.1:5760` for the GCS view.

REQUIREMENTS

Linux or WSL (Ubuntu 22.04+). The install script handles all dependencies — Python, gcc, pymavlink, MAVProxy. Takes ~10 minutes first time.

Three commands, and you have a simulated aircraft. **Every mission you test in SITL is a mission you do not crash in real life.**